

# Ch1 Manifolds

## Differentiable Manifolds

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## Notations

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# Notations (I)

- **injective:** A map  $f: A \rightarrow B$ ; if  $\forall x, y \in A, x \neq y \Rightarrow f(x) \neq f(y)$ , then  $f$  is an *injective map*.
- **surjective:** A map  $f: A \rightarrow B$  is a *surjective map* if  $f$  is onto, i.e.  $f(A) = B$ .
- A *commutative diagram* (of maps)

$$\begin{array}{ccc} & A & \\ f \swarrow & & \searrow h \\ B & \xrightarrow{g} & C \end{array} \quad \text{means} \quad h = g \circ f.$$

- $\mathbb{R}^d$  —  $d$ -dim Euclidean space.

## Notations (II)

- The function  $r_i : \mathbb{R}^d \rightarrow \mathbb{R}$ ,  $(x_1, \dots, x_d) \mapsto x_i$ , is called the  $i$ -th (canonical) coordinate function on  $\mathbb{R}^d$ .  
If  $f : X \rightarrow \mathbb{R}^d$ , then  $f^i := r_i \circ f$  is the  $i$ -th component function of  $f$ .
- $B_p(r)$  — the open ball of radius  $r$  about  $p$ .
- Given a function  $f : X \rightarrow \mathbb{R}$ ,

$$\text{supp } f := \overline{f^{-1}(\mathbb{R} \setminus \{0\})}$$

is called the *support* of  $f$ .

- Given a multi-index  $\alpha$ :

$$\frac{\partial^\alpha}{\partial x^\alpha} := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}.$$

# Differentiable Manifolds

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## Definition ( $C^k$ functions)

Let  $U \subset \mathbb{R}^d$  be open,  $f: U \rightarrow \mathbb{R}$ . We say  $f$  is *differentiable of class  $C^k$*  on  $U$  (for  $k \geq 0$ ) if  $\frac{\partial^\alpha}{\partial x^\alpha} f$  exists and is continuous for all  $|\alpha| \leq k$ .

- In particular,  $f$  is  $C^0$  if  $f$  is continuous.
- $f: U \rightarrow \mathbb{R}^n$  is  $C^k$  if each component of  $f$  is  $C^k$  on  $U$ .
- $f$  is  $C^\infty$  if  $f$  is  $C^k$  for all  $k \geq 0$ .

# Locally Euclidean spaces

## Definition (Locally Euclidean space)

A *locally Euclidean space*  $M$  of  $\dim d$  is a Hausdorff topological space  $M$  for which each point  $p \in M$  has a neighborhood homeomorphic to an open subset of  $\mathbb{R}^d$ .

- If  $\varphi$  is a homeomorphism of a connected open set  $U \subset M$  to an open subset of  $\mathbb{R}^d$ , then  $\varphi$  is called a *coordinate map*;
- $x_i := r_i \circ \varphi$  is called the *coordinate function*;
- $(U, \varphi)$ , i.e.  $(U, (x_1, \dots, x_d))$ , is called a *coordinate system*;
- if  $p \in U$  s.t.  $\varphi(p) = 0$ , the coordinate system is said to be *centered at  $p$* .

# Differentiable structure

## Definition (Differentiable structure)

A *differentiable structure*  $\mathcal{F}$  of class  $C^k$  ( $1 \leq k \leq \infty$ ) on a locally Euclidean space  $M$  is a collection of coordinate systems  $\{(U_\alpha, \varphi_\alpha) \mid \alpha \in A\}$  satisfying:

- (a) the coordinate systems *cover*  $M$ :  $\bigcup_{\alpha \in A} U_\alpha = M$ ;
- (b) whenever  $U_\alpha \cap U_\beta \neq \emptyset$ , the map  $\varphi_\alpha \circ \varphi_\beta^{-1}$  defined on the image of  $U_\alpha \cap U_\beta$  is of class  $C^k$ ,  $\forall \alpha, \beta \in A$ ;
- (c) the collection  $\mathcal{F}$  is *maximal*: if  $(U, \varphi)$  is a coordinate system s.t.  $\varphi_\alpha \circ \varphi^{-1}$  and  $\varphi \circ \varphi_\alpha^{-1}$  are  $C^k$  for all  $\alpha \in A$ , then  $(U, \varphi) \in \mathcal{F}$ .

# Differentiable manifolds

## Definition (Differentiable manifold)

A  $d$ -dimensional differentiable manifold of class  $C^k$  is a pair  $(M, \mathcal{F})$  consisting of a second countable, locally Euclidean space  $M$  of dim  $d$ , together with a differentiable structure  $\mathcal{F}$  of class  $C^k$ .

## Remark

If  $\mathcal{F}_0 = \{(U_\alpha, \varphi_\alpha) \mid \alpha \in A\}$  is any collection of local coordinate systems satisfying (a) & (b), there is a *unique maximal* collection containing  $\mathcal{F}_0$ . Namely

$$\mathcal{F} = \{(U, \varphi) \mid \varphi \circ \varphi_\alpha^{-1} \text{ \& } \varphi_\alpha \circ \varphi^{-1} \text{ are } C^k \text{ for all } (U_\alpha, \varphi_\alpha) \in \mathcal{F}_0\}.$$

## Exercise

Prove  $\mathcal{F}$  satisfies (a) & (b), as well as (c).

## Remark (other structures)

There are differentiable structures of class  $C^\omega$  and *complex analytic* structures:  $C^\omega$  refers to the composition  $\varphi_\alpha \circ \varphi_\beta^{-1}$  being given by convergent power series; a complex analytic structure refers to the composition given by holomorphic functions.

## Remark

In this lecture, we only focus on the differentiable structure of class  $C^\infty$ .

## Examples of Differentiable Manifolds

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## Examples (a), (b)

### Example

$\mathbb{R}^d$  is a manifold. A differentiable structure  $\mathcal{F}$  is the *maximal collection* of coordinate systems containing  $(\mathbb{R}^d, \text{Id})$ .

### Example

A *real* vector space  $V$  has a natural manifold structure. Let  $\{e_i\}$  be a basis of  $V$  and  $\{f_i\}$  the dual basis of  $V^*$ , i.e.  $f_i(e_j) = \delta_{ij}$ . Then  $(V, (f_1, \dots, f_n))$  is a coordinate system.

## Example (c): the $d$ -sphere

### Example (The $d$ -sphere)

$S^d = \{(x_1, \dots, x_{d+1}) \mid x_1^2 + \dots + x_{d+1}^2 = 1\}$  is a manifold. Denote

$$n = (0, \dots, 0, 1), \quad s = (0, \dots, 0, -1).$$

Then  $(S^d \setminus \{n\}, \pi_n)$ ,  $(S^d \setminus \{s\}, \pi_s)$  are *local coordinates* (stereographic projections) contained in its differentiable structure  $\mathcal{F}$ .

### Exercise

Verify that  $\pi_n = \varphi_n^{-1}$ , where

$$\varphi_n(u) = \left( \frac{2u}{1 + \|u\|^2}, \frac{\|u\|^2 - 1}{1 + \|u\|^2} \right) \in S^d, \quad u = (u_1, \dots, u_d) \in \mathbb{R}^d.$$

## Examples (d), (e)

### Example

An open subset  $U$  of a differentiable manifold  $(M, \mathcal{F}_M)$  is itself a differentiable manifold with

$$\mathcal{F}_U = \{ (U_\alpha \cap U, \varphi_\alpha|_{U_\alpha \cap U}) \mid (U_\alpha, \varphi_\alpha) \in \mathcal{F}_M \}.$$

### Example (The general linear group)

$\mathrm{GL}_n(\mathbb{R}) \subset \mathbb{R}^{n^2}$  is an *open subset* of  $\mathbb{R}^{n^2}$ , therefore it is a manifold — since  $\det : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$  is continuous and

$$\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\}).$$

## Example (f): product manifolds

### Example (Product manifolds)

Let  $(M_1, \mathcal{F}_1)$  and  $(M_2, \mathcal{F}_2)$  be differentiable manifolds of  $\dim d_1, d_2$ . Then  $M_1 \times M_2$  becomes a differentiable manifold of  $\dim d_1 + d_2$ , with the differentiable structure  $\mathcal{F}$  the *maximal collection* containing

$$\{(U_\alpha \times V_\beta, (\varphi_\alpha, \psi_\beta)) \mid (U_\alpha, \varphi_\alpha) \in \mathcal{F}_1, (V_\beta, \psi_\beta) \in \mathcal{F}_2\}.$$

## Example (g): projective space $\mathbb{RP}^n$

### Example (Projective space)

Define  $\mathbb{RP}^n$  to be the *quotient topological space* of  $\mathbb{R}^{n+1} \setminus \{0\}$ :

$$\mathbb{RP}^n = (\mathbb{R}^{n+1} \setminus \{0\}) / \sim,$$

with “ $\sim$ ” the equivalence relation

$$X \sim Y \iff \exists \lambda \neq 0 \text{ s.t. } X = \lambda Y.$$

Introducing *homogeneous coordinates* to denote points of  $\mathbb{RP}^n$ :

$$[x] := (x_1 : x_2 : \cdots : x_{n+1}).$$

## Example (g): local coordinates on $\mathbb{RP}^n$

### Example (Projective space (continued))

Consider the open set

$$U_k = \{x \in \mathbb{R}^{n+1} \mid x_k \neq 0\} / \sim \subset \mathbb{RP}^n,$$

and denote

$$\varphi_k([x]) = \left( \frac{x_1}{x_k}, \dots, \frac{\widehat{x_k}}{x_k}, \dots, \frac{x_{n+1}}{x_k} \right) \in \mathbb{R}^n.$$

Then  $(U_k, \varphi_k)$  gives a local coordinate system of  $\mathbb{RP}^n$ .

$\Rightarrow \mathbb{RP}^n$  is a differentiable manifold with differentiable structure  $\mathcal{F}$  containing

$$\{(U_k, \varphi_k) \mid k = 1, \dots, n+1\}.$$

# Smooth Functions and Maps

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# $C^\infty$ functions and $C^\infty$ maps

## Definition ( $C^\infty$ functions and $C^\infty$ maps)

We say  $f: M \rightarrow \mathbb{R}$  is a  $C^\infty$  *function* on  $M$  if  $f \circ \varphi_\alpha^{-1}$  is  $C^\infty$  for each coordinate map  $(U_\alpha, \varphi_\alpha)$  on  $M$ .

A continuous map  $f: M \rightarrow N$  is a  $C^\infty$  *map* between manifolds  $M$  and  $N$  if

$$\psi_\beta \circ f \circ \varphi_\alpha^{-1} \in C^\infty \quad \text{for all } (U_\alpha, \varphi_\alpha) \text{ on } M \text{ and } (V_\beta, \psi_\beta) \text{ on } N.$$